Light

GCSE Computer Science — FDE Cycle and Systems (Set 3)

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GCSE • OCR J277 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (OCR J277)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

Before the CPU can process any instruction, what must happen first?

- ○ **A.** The result must be displayed on screen
- ○ **B.** The user must press a key
- ○ **C.** The instruction must be fetched from memory
- ○ **D.** The hard drive must spin up

Q2 (1 mark)

What is determined during the decode stage of the FDE cycle?

- ○ **A.** What operation the instruction requires the CPU to perform

- ○ **B.** The colour of the display output
- ○ **C.** How much the program costs
- ○ **D.** The name of the file being processed

Q3 (1 mark)

At what point in the FDE cycle is the instruction actually carried out?

- ○ **A.** During the fetch stage
- ○ **B.** During the execute stage
- ○ **C.** During the decode stage
- ○ **D.** Before the cycle begins

Q4 (1 mark)

The Program Counter contains the value 500. What does this tell the CPU?

- ○ **A.** The program has 500 lines of code
- ○ **B.** 500 seconds have passed since the program started
- ○ **C.** The next instruction to fetch is at memory address 500
- ○ **D.** The CPU is running at 500 MHz

Q5 (1 mark)

When the CPU needs to read data from memory location 750, which register holds this address?

- ○ **A.** The Memory Address Register (MAR)
- ○ **B.** The Accumulator
- ○ **C.** The Memory Data Register (MDR)
- ○ **D.** The Current Instruction Register (CIR)

Q6 (1 mark)

During the fetch stage, the instruction retrieved from memory is placed into which register?

- ○ **A.** The Program Counter
- ○ **B.** The Memory Address Register
- ○ **C.** The Memory Data Register (MDR)
- ○ **D.** The Arithmetic Logic Unit

Q7 (1 mark)

Before the ALU can perform an addition, it needs input values. Where might one of these values come from?

- ○ **A.** The monitor
- ○ **B.** The printer
- ○ **C.** The keyboard

- ○ **D.** The Accumulator

Q8 (1 mark)

A smart thermostat monitors room temperature and controls the heating system. What type of system is this?

- ○ **A.** A gaming console
- ○ **B.** An embedded system
- ○ **C.** A supercomputer
- ○ **D.** A web server

Q9 (1 mark)

A digital camera contains processing hardware to capture and store images. This is an example of:

- ○ **A.** A mainframe computer
- ○ **B.** A desktop workstation
- ○ **C.** A cloud server
- ○ **D.** An embedded system

Q10 (1 mark)

After one FDE cycle completes, what happens next?

- ○ **A.** The computer shuts down

- ○ **B.** Another FDE cycle begins for the next instruction
- ○ **C.** The user must click 'continue'
- ○ **D.** All data is saved to the hard drive

Q11 (1 mark)

Embedded systems typically cost less to produce than general-purpose computers with similar processing power. Why?

- ○ **A.** Embedded systems use more expensive components
- ○ **B.** General-purpose computers are simpler to manufacture
- ○ **C.** Embedded systems always use the latest technology
- ○ **D.** They only include the components necessary for their specific function, avoiding unnecessary features

Q12 (1 mark)

The Control Unit sends signals to other CPU components. What do these signals coordinate?

- ○ **A.** The colour of the computer case
- ○ **B.** The timing and operation of data movement and processing within the CPU
- ○ **C.** The price of electricity
- ○ **D.** The speed of the internet connection

Q13 (2 marks)

Describe the role of the address bus during the fetch stage.

- ○ **A.** It carries the fetched instruction back to the CPU
- ○ **B.** It stores instructions while they wait to be processed
- ○ **C.** It supplies power to the memory chips
- ○ **D.** It carries the memory address from the MAR to memory, specifying which location to access

Q14 (2 marks)

Instructions are stored in memory as binary patterns. Explain what happens when the Control Unit decodes an instruction.

- ○ **A.** The binary is converted to audio and played through speakers
- ○ **B.** The binary pattern is interpreted to identify the operation type and prepare appropriate control signals
- ○ **C.** The Control Unit ignores the binary and guesses the operation
- ○ **D.** The binary is deleted from memory after decoding

Q15 (2 marks)

A CMP (compare) instruction compares two values during the execute stage.
What happens as a result of this comparison?

- ○ **A.** Status flags are set to indicate the comparison result (equal, greater, less), which can be checked by subsequent instructions

- ○ **B.** The two values are added together
- ○ **C.** Both values are deleted from memory
- ○ **D.** The comparison result is printed on paper

Q16 (2 marks)

An elevator (lift) contains an embedded control system.
Explain two functions this system must perform.

- ○ **A.** Playing video games and browsing social media
- ○ **B.** Editing photographs and creating documents
- ○ **C.** Processing floor requests from buttons and controlling motor operation to move between floors safely
- ○ **D.** Hosting websites and managing email

Q17 (2 marks)

When a branch instruction modifies the Program Counter, what effect does this have on program execution?

- ○ **A.** The program continues with the next sequential instruction as normal
- ○ **B.** All registers are reset to zero
- ○ **C.** The program immediately terminates
- ○ **D.** Execution continues from a different memory location instead of the next sequential address

Q18 (2 marks)

An ATM (cash machine) uses embedded systems for its operation. Identify why embedded characteristics are important for this application.

- ○ **A.** ATMs need to run general office software
- ○ **B.** ATMs require the ability to play music
- ○ **C.** Reliability ensures continuous availability, dedicated function prevents misuse, and real-time response provides immediate transaction feedback
- ○ **D.** Users want to install their own applications

Q19 (2 marks)

CPU registers, cache memory, and RAM form a memory hierarchy. Explain why registers are at the top of this hierarchy.

- ○ **A.** Registers store the most data
- ○ **B.** Registers provide the fastest access because they are inside the CPU, enabling immediate data availability during instruction execution
- ○ **C.** Registers are the slowest memory type
- ○ **D.** Registers are the cheapest to manufacture

Q20 (2 marks)

The data bus is bidirectional, meaning data can flow in both directions. Explain why this is necessary for CPU operation.

- ○ **A.** Bidirectional flow is actually not needed

- ○ **B.** The CPU only ever reads data, never writes
- ○ **C.** The CPU must both read data from memory (during LOAD operations and instruction fetch) and write data to memory (during STORE operations)
- ○ **D.** Data flows in one direction only

Q21 (2 marks)

A drone uses embedded systems to maintain stable flight.
Describe how sensor input enables flight stability.

- ○ **A.** Accelerometers and gyroscopes detect orientation changes, and the embedded controller adjusts motor speeds in real-time to maintain level flight
- ○ **B.** Sensors are not used in drones
- ○ **C.** The drone asks the user which direction to fly every second
- ○ **D.** Stability is achieved by making the drone very heavy

Q22 (2 marks)

A MUL (multiply) instruction requires more processing than a simple ADD instruction.
What is the typical consequence of this during execution?

- ○ **A.** MUL executes faster than ADD
- ○ **B.** MUL and ADD always take exactly the same time
- ○ **C.** MUL requires more clock cycles to complete than ADD

○ **D.** MUL doesn't use the ALU at all

Q23 (2 marks)

Smart electricity meters contain embedded systems that communicate usage data to suppliers.
Explain why embedded systems are suitable for this application.

○ **A.** Continuous monitoring with low power consumption, reliable long-term operation without maintenance, and secure data transmission are essential

○ **B.** They need to play video content

○ **C.** Users want to program the meters themselves

○ **D.** Meters require graphics-intensive processing

Q24 (3 marks)

Trace through the complete FDE cycle for an INC (increment) instruction that adds 1 to the value in the Accumulator. The instruction is at memory address 80.
Which sequence correctly describes this process?

○ **A.** The increment happens during fetch, before decode

○ **B.** The Accumulator is never involved in INC instructions

○ **C.** INC instructions don't use the ALU

○ **D.** Fetch: PC (value 80) → MAR, instruction retrieved to MDR then CIR, PC becomes 81. Decode: CU identifies INC instruction targeting Accumulator. Execute: ALU adds 1 to Accumulator value, result stored back in Accumulator

Q25 (3 marks)

A car's Engine Control Unit (ECU) is an embedded system that manages fuel injection, ignition timing, and emissions control.
Analyse why embedded characteristics are critical for this application.

- ○ **A.** Real-time response adjusts engine parameters thousands of times per second; reliability ensures consistent operation over the vehicle's lifetime; harsh environment resistance (heat, vibration) requires robust embedded design
- ○ **B.** The ECU needs to display high-resolution graphics
- ○ **C.** The ECU runs spreadsheet software
- ○ **D.** Passengers need to reprogram the ECU while driving

Q26 (3 marks)

A 32-bit CPU has registers and data buses that handle 32 bits at a time.
Analyse how this affects the FDE cycle compared to a 16-bit CPU.

- ○ **A.** The number of bits has no effect on CPU operation
- ○ **B.** 16-bit CPUs are faster because they have fewer bits to process
- ○ **C.** 32-bit CPUs fetch and process twice as many bits per cycle as 16-bit CPUs; this means larger instructions and data values can be handled in single operations, reducing the number of cycles needed for complex tasks
- ○ **D.** 32-bit CPUs cannot run the FDE cycle at all

Q27 (3 marks)

Cache memory stores frequently accessed data close to the CPU. A 'cache miss' occurs when needed data isn't in the cache.
Analyse how cache misses affect the FDE cycle.

- ○ **A.** When a cache miss occurs during fetch or execute, the CPU must wait for slower main memory access, potentially stalling the pipeline for many clock cycles while data is retrieved from RAM
- ○ **B.** Cache misses speed up execution
- ○ **C.** Cache misses have no effect on the FDE cycle
- ○ **D.** Cache misses only affect the decode stage

Q28 (3 marks)

A patient monitoring system in a hospital tracks vital signs and alerts staff to abnormalities.
Analyse the consequences if this embedded system failed to provide real-time response.

- ○ **A.** There would be no consequences
- ○ **B.** Staff would need to use faster internet
- ○ **C.** The only consequence would be slightly higher electricity bills
- ○ **D.** Delayed detection of life-threatening conditions (cardiac arrest, respiratory failure) could result in patient death; late alerts prevent timely intervention; degraded continuous monitoring could miss developing problems

Q29 (3 marks)

A HALT instruction tells the CPU to stop executing instructions.
Analyse what happens when HALT is executed and how program execution can resume.

- ○ **A.** HALT execution stops the FDE cycle from continuing; the CPU waits in a low-power state until an external event (interrupt, reset, or

power cycle) causes execution to resume; the PC is not incremented after HALT

○ **B.** HALT has the same effect as any other instruction — the next instruction executes normally

○ **C.** HALT permanently destroys the CPU

○ **D.** HALT speeds up the next instruction

Q30 (3 marks)

Autonomous vehicles contain numerous embedded systems for sensor processing, decision making, and vehicle control.
Analyse why multiple embedded systems are used rather than a single central computer.

○ **A.** Using multiple systems is just to increase cost

○ **B.** Specialised systems optimise for their specific function (image processing for cameras, real-time control for braking); isolation prevents failures in one system from affecting others; redundancy enables continued operation if one system fails; parallel processing handles the massive data flow from dozens of sensors simultaneously

○ **C.** A single computer would actually be more reliable

○ **D.** Multiple systems are used because programmers prefer it

Total: **55** marks • Time: **30 min**

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